

**Discuss your approach to the anaesthetic management of a screaming 5-year-old child with penetrating eye injury.**

(Modified from FHKCA 2009A, FANZCA 2015A)

Examiner Report

*7. What are the priorities in the initial evaluation and management of this child? What further information would you need prior to taking him to theatre? (3/18 passed)*

This case involves a minor where no history is available from the patient himself. There is circumstantial evidence of significant trauma with the added issue of possible burns +/- inhalational injury. Candidates are therefore expected to follow a systematic approach to avoid missing any injury and management priorities should be according to the ATLS guidelines. In particular, they should look for injuries which children are prone to yet signs are subtle: i.e. head injury and chest trauma. Moreover, they are expected to be on the lookout for airway burns and give airway management a high priority. Pain relief should be offered once the child is stabilized and candidates are expected to discuss with their surgical colleagues to optimize the timing of surgery to salvage the patient's eye.

Candidates were expected to mention the following

- A. Recognise this is an acute case and problems associated with that / often can wait      Considers potential for loss of contents of eye and strategies to manage that and      considers mechanism of injury and its implications
- B. Should demonstrate that they are making an attempt to minimise the risk of increasing IOP, particularly at induction and emergence - but also intra-operatively. Candidates should demonstrate due regard for the potentially non fasted patient Note that the question asks them to justify their anaesthetic technique so purely describing a GA is inadequate.

## Sample Answer

This is an uncooperative pre-school patient who requires an urgent eye operation. He is likely in pain, and may not be well-fasted. It is important to avoid situations and drugs which may increase intra-ocular pressure (IOP).

### **Pre-op**

- **I will speak to surgeon regarding salvageability of the eye and urgency of the operation.**
  - If eye is not salvageable, I will wait for adequate fasting time (6h from last meal) to minimise aspiration risk.
  - Analgesia should be provided in the mean time.
  - Prior to sending for the patient (to OT), I will request for ward staff to apply Ametop cream covered with a clear, adhesive dressing over bilateral hand dorsum.
- **I will speak to patient and parents to get a targeted history to confirm that this is an isolated injury.**
  - Mechanism of injury? Fasting time?
  - Any significant trauma? Concomitant injuries?
  - Other past medical history, including any drug allergy, recent URTI, family history of problems with general anaesthesia
- **If patient is inconsolable during my assessment at the reception,**
  - I will pre-medicate him with PO Midazolam 0.5mg/kg.
  - This will minimise patient's distress (which could increase IOP), and facilitate patient's cooperation during induction. This is balanced against aspiration risk.
  - I will also request for parental presence during induction, with distraction with toys/games/video (if the other eye is intact).

### **Intra-op**

- Assuming child is sedated, with adequate time for Ametop cream application and IV cannulation, my preferred approach is that of IV induction (modified RSI) with micro-cuff ETT.
  - Pre-oxygenation with 100% O<sub>2</sub>, with parental presence, in warmed OT.
  - Induction with IV Propofol 2-3mg/kg, IV Rocuronium 1mg/kg, with standby IV Sugammadex 16mg/kg for reversal.
  - IV Lignocaine 1mg/kg and IV Fentanyl 2mg/kg to blunt cardiovascular response to laryngoscopy
  - Decompress stomach with OG tube after intubation
- **I will take careful measures to avoid worsening IOP intra-op.**
  - Maintain adequate depth of anaesthesia and muscle relaxation to avoid coughing/straining
  - Titrate ventilation settings to avoid hypoxia, hypercarbia.
  - Multi-modal analgesia to treat pain/ nociception
    - Paracetamol and NSAIDS (if not already given in the ward/ED)
    - Short acting opioids (Fentanyl)
    - Topical LA/ RA by surgeon.
  - Avoid drugs which may increase IOP
    - Nitrous Oxide
    - Suxamethonium (controversial)
    - Ketamine (controversial)
- **I will monitor and prepare for oculo-cardiac and oculo-emetic reflexes intra-op.**
  - IV Atropine 20mcg/kg drawn and ready for administration. Inform surgeon to stop if patient becomes severely bradycardic during manipulation.
  - Give anti-emetics IV Dexamethasone 0.1mg/kg and IV Ondansetron 0.15mg/kg.

- I will extubate patient awake in a smooth and controlled manner.
  - Slow and gentle washing out of inhalational agent
  - Ensuring adequate analgesia before extubation
  - Anti-emetic prophylaxis (IV Dexamethasone 0.1mg/kg and IV Ondansetron 0.15mg/kg)
  - IV Lignocaine 1mg/kg to blunt cardiovascular response during extubation
  - Minimize contact and noise stimulation in OT

#### **Post-op**

- I will recover patient in a quiet and dark area, with parental presence, in PACU.
  - This is to minimise emergence delirium, which patient is at higher risk of due to his age, significant distress pre-op, impaired vision (eye shield).
- To general ward if nil complications arising from surgery or anaesthesia.
- Regular analgesia with Paracetamol + anti-emetics as necessary.